

Identifying Factors that Limit Discriminative EEG Task Performance

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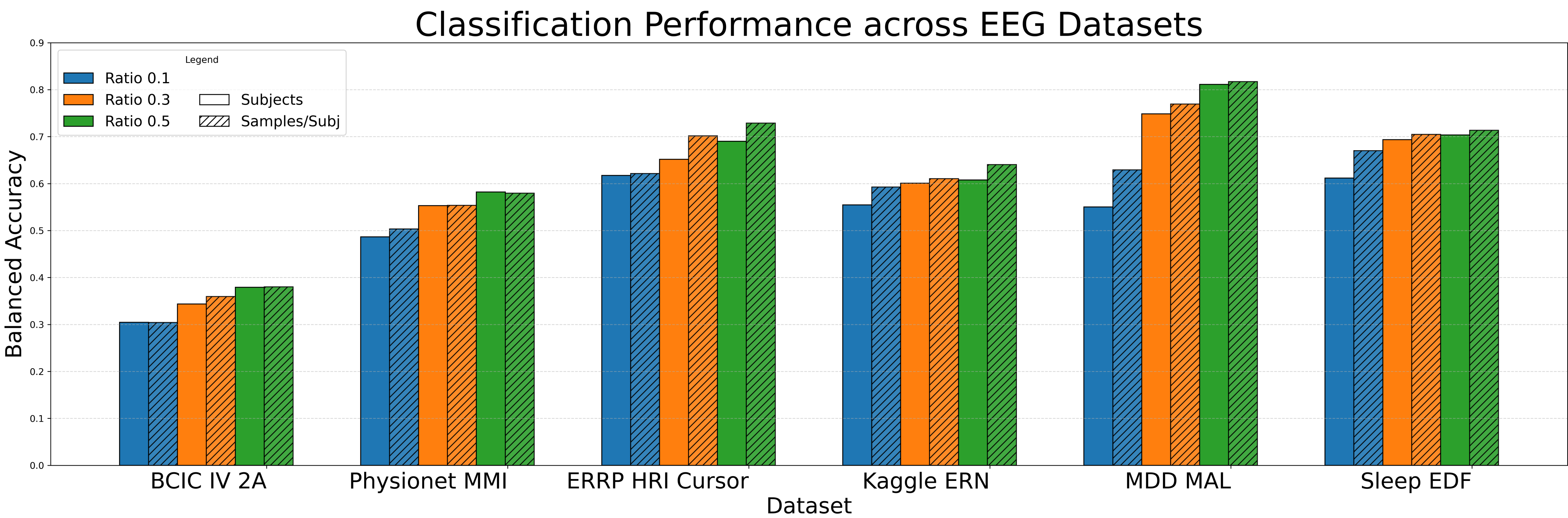


Introduction

- Electroencephalography (EEG) is a non-invasive biomedical imaging modality with varied uses, such as in Epilepsy Monitoring Units, ICUs, Sleep Disorder Monitoring in clinical contexts
- EEG is also employed in understanding the underlying cognitive processes during affective, visual and auditory processing
- EEG as a modality is **noisy**, exhibits **high inter** and **intra** subject variability and is **time-intensive** to setup for experiments

Motivation

- In most clinical settings, ability to collect data is constrained owing to limitations in resources
- For EEG, number of EEG channels in montage and number of trials in experiment significantly impact duration of experiment
- The ability to identify critical factors that impact downstream performance could not only help deploy models in resource constrained settings, but also guide data collection



Datasets and Models

Dataset	Channels	Classes	Task
BCIC IV 2A	22	4	BCI
Physionet MMI	64	4	BCI
ERRP HRI	27	2	ERN
Kaggle ERN	56	2	ERN
MDD MAL	19	2	Mental Health
Sleep EDF	2	5	Sleep Stage

- Results are shown for supervised EEG models: EEGNet, SparcNet and ContraWR
- All models are trained across multiple seeds

Results

- Preliminary experiments varying ratio of total subjects and proportion of samples per subject in training set indicate diversity of subjects is more critical for performance compared to the proportion of samples per subject
- The importance of participant diversity is observed only when operating in very low data regimes i.e. 10% of training data
- ERP based tasks (BCIC IV 2A, Physionet MMI, Kaggle ERN) are less dependent on sample diversity indicating inter-subject variability impacts longer window tasks

Ongoing Experiments

- To provide generalization on findings, experiments are being run on linear probing with pre-trained EEG SSL models